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MVJ22A6073 -IOT - LEVEL 3

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A PROJECT REPORT ON

“FLORAGUARD”

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

IN

IOT ENGINEERING

Submitted By

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MVJ COLLEGE OF ENGINEERING

BANGALORE - 67

ACADEMIC YEAR

2026- 2027



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CERTIFICATE

This is to certify that the IOT Level 3 project work titled “**Floraguard**” is carried out by us who are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2026 - 2027. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report deposited in the departmental library. The project report has been approved as satisfies the academic requirements in respect of major project work prescribed by the institution for the said Degree.

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EXTERNAL EXAMINERS

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DECLARATION

We, students of Sixth semester B.E., Department of Electronics and Communication Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the IOT Level 3 project titled “**Floraguard**” has been carried out by us and submitted in partial fulfilment for the award of Degree of Bachelor of Engineering in Electronics and Communication Engineering during the year 2026 - 2027. Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

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ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany a successful completion of any task would be incomplete without the mention of people who made it possible, success is the epitome of hard work and perseverance, but steadfast of all is encouraging guidance.

So, with gratitude we acknowledge all those whose guidance and encouragement served as beacon of light and crowned our effort with success.

We are thankful to **Dr. Ajayan K R, Principal of MVJCE**, for his encouragement and support throughout the project work.

We are also thankful to **Dr. M Thilagaraj, HOD, Department of IIOT**, for his incessant encouragement & all the help during the project work.

We consider it a privilege and honor to express our sincere gratitude to our guides **Dr. MANU RAJU Assistant Professor, ECE Department** and **Prof. SACHIN T Assistant Professor, ECE(ACT) Department** for their valuable guidance throughout the tenure of this project work, and whose support and encouragement made this work possible.

It is also an immense pleasure to express our deepest gratitude to all faculty members of our department for their cooperation and constructive criticism offered, which helped us a lot during our project work.

Finally, we would like to thank all our family members and friends for whose encouragement and support was invaluable.

Thanking You

Floraguard is a virtual plant caretaker designed to simplify and enhance home gardening through intelligent monitoring, personalized care guidance, and real-time decision support. Many home gardeners struggle with identifying plant health issues early, tracking proper care routines, and maintaining consistent growth records. **Floraguard** addresses these challenges by integrating image based plant disease detection, an AI-driven care planner, and an interactive growth timeline into a single, user-friendly platform. Beyond these core features, **Floraguard** promotes an interactive gardening experience with reminders, environmental tips, carbon footprint tracking, and motivational insights that emotionally connect users with their plants. By integrating technology with home gardening, **Floraguard** aims to make plant care more intuitive, reliable, and enjoyable. The project ultimately demonstrates how AI, computer vision, and data-driven recommendations can empower individuals regardless of gardening experience to nurture healthier plants, reduce errors, and build a more sustainable relationship with nature.

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CHAPTER 1

INTRODUCTION

Home gardening has grown significantly in recent years as people seek sustainable living practices, emotional comfort, and a closer connection with nature. However, despite its increasing popularity, many home gardeners—especially beginners—struggle with maintaining plant health consistently. Issues such as late detection of diseases, improper watering cycles, incorrect sunlight exposure, and lack of timely fertilization often lead to poor plant growth or plant loss. Traditional care methods rely heavily on manual observation and experience, making plant maintenance challenging for individuals with limited knowledge or time. To address these gaps, technology-driven solutions have emerged, providing digital assistance to simplify gardening tasks and promote healthier plant care practices.

Floraguard is designed as a comprehensive virtual plant caretaker that blends artificial intelligence, computer vision, and personalized care guidance into a single platform. The system's core objective is to empower home gardeners by providing intelligent tools that reduce uncertainty and make plant care more intuitive. One of the major components of **Floraguard** is its plant disease detection module, which uses image analysis to identify common plant diseases at an early stage. By alerting users promptly and suggesting remedial measures, the system helps

prevent disease spread and reduces the chances of irreversible damage. This early detection capability is particularly valuable for beginners who may not recognize subtle symptoms of plant stress.

The project also incorporates supportive tools like carbon-footprint tracking, environmental tips, and customizable reminders, strengthening the user's overall gardening experience. By merging technical innovation with the emotional aspects of home gardening, **Floraguard** aims to make plant care accessible, efficient, and enjoyable. Ultimately, the system showcases how AI-driven applications can transform everyday activities, encourage sustainable practices, and support individuals in building a deeper, more informed relationship with their plants.

CHAPTER 2

PROBLEM STATEMENT

Home gardening has become an increasingly popular activity, yet many individuals especially beginners and busy households struggle to maintain healthy and thriving plants. Plant care requires consistent attention, accurate observation, and proper knowledge of plant-specific needs. However, most home gardeners lack the expertise to identify early symptoms of diseases, understand changing environmental requirements, and maintain structured care routines. As a result, plants often suffer from preventable issues such as fungal infections, nutrient deficiencies, improper watering, and inconsistent sunlight exposure.

Traditional methods of plant care rely heavily on manual inspection and personal experience, making them unreliable for novices. Early signs of plant diseases are subtle and easily overlooked, leading to delayed treatment when the damage has already escalated. Additionally, gardeners often struggle to remember routine care schedules, causing overwatering, underwatering, or irregular fertilization. Keeping track of plant growth stages, seasonal adjustments, and environmental conditions adds further complexity. There is a clear need for a

systematic, user-friendly solution that can guide individuals in their daily plant care activities while reducing dependency on expert knowledge.

Therefore, there is a pressing need for an intelligent virtual plant caretaker that can automate disease detection, provide customized care recommendations, track growth timelines, and support sustainable gardening practices. Such a system should simplify plant maintenance, reduce the risk of plant loss, and empower individuals regardless of expertise to cultivate healthy plants with confidence. Floraguard aims to address these challenges by offering an integrated, AI-assisted platform that transforms the way people manage and connect with their home gardens.

CHAPTER 3

LITERATURE SURVEY

2.1 Agri Doctor: Intelligent Crop Disease Detection for Sustainable Farming

AUTHORS: Not specified

PUBLISHED ON: 2025

This paper introduces “Agri Doctor” a next-generation crop disease detection system aimed at improving sustainable farming outcomes. The system integrates deep-learning-based plant disease analysis with environmental and soil data to create a holistic digital advisor for farmers. A high-resolution crop image data set is collected using mobile devices, drones, and low-cost field cameras. A custom Convolutional Neural Network (CNN) model is trained to classify major diseases with high accuracy, while also identifying nutrient deficiencies based on leaf discoloration and texture changes. The system incorporates weather forecasts, soil moisture

readings, and fertilizer history to offer context-aware recommendations. The study demonstrates that real-time disease detection and predictive analytics reduce crop loss, support precision agriculture, and empower farmers with actionable insights delivered through a mobile app. Agri Doctor highlights how AI-driven tools can significantly enhance decision-making and productivity in resource-constrained rural environments.

2.2 Enhanced Rice Leaf Diseases Diagnosis Using Deep Learning Model

GAN-Based Synthetic Data Augmentation

AUTHORS: J. Dhanasekar; M. Ishwarya Niranjana; G. Suresh; R. Thirugnanabharathi; K. Gokul Parasath

PUBLISHED ON: 2025

This research focuses on overcoming the challenge of limited rice disease datasets by integrating Generative Adversarial Networks (GANs) to generate realistic synthetic leaf images. The authors create an enriched data set that includes variations in lighting, background noise, disease severity levels, and leaf orientation. A hybrid deep-learning architecture composed of Efficient Net and Dense Net feature extractors is trained on both real and GAN-generated images. The system classifies diseases such as leaf blast, bacterial blight, and brown spot with significantly improved accuracy. Ablation studies prove that the GAN-augmented data set improves model performance by addressing class imbalance and reducing over fitting. The proposed system is intended for deployment on mobile platforms for field-level diagnosis.

2.3 A Novel Approach for Generic Plant Disease Classification

AUTHORS: Puneet Mugabasav, Shreyas M. Salotagi, Kulkarni Akash.

PUBLISHED ON: 2025

This study proposes a universal disease classification model capable of identifying diseases across multiple plant species using a single framework. The authors construct a large-scale, crossspecies dataset combining leaves of tomato, potato, apple, maize, pepper, and other crops. A novel architecture combining attention mechanisms with EfficientNet-B4 is introduced to capture fine-grained disease patterns. The system leverages transfer learning and domain adaptation to generalize across variable leaf structures, textures, and lighting conditions. Experiments show high robustness to field-level noise such as shadows, background clutter, and partial leaf images. The work highlights the importance of generalized models for scalable, realworld agricultural

applications. The system significantly reduces the need for species-specific models and simplifies digital farming workflows.

2.4 Advancing Agricultural Health: Hybrid CNN–SVM Framework for Classifying Tomato Diseases

AUTHORS: J. Dhanasekar, M. Ishwarya Niranjana, G. Suresh, R. Thirugnanabharathi, K. Gokul Prasath

PUBLISHED ON: 2023

This paper presents a hybrid approach that merges the feature extraction strength of Convolutional Neural Networks with the classification precision of Support Vector Machines. The CNN portion extracts deep spatial features from tomato leaf images, focusing on color patterns, lesion shapes, and progression markers. These features are fed into an SVM classifier with an RBF kernel for improved boundary separation between similar disease categories. The hybrid model addresses the limitations of CNNs in distinguishing visually close diseases like early blight and late blight. Comparative experiments reveal that the CNN-SVM model offers higher accuracy and faster convergence than standalone CNN models. The study concludes that combining deep and classical machine learning methods can yield highly efficient and reliable agricultural diagnostic systems.

2.5 Comparative Analysis of Different Plant Leaf Disease Classification Detection Using CNN

AUTHORS: Nilima Gadkar, Subhranil Das, Sarabani Chakraborty, Sudhansu Kumar Mishra

PUBLISHED ON: 2024

This comparative paper evaluates multiple CNN architectures—VGG16, InceptionV3, ResNet50, Exception, DenseNet201, and MobileNetV2—for plant disease detection. Each architecture is tested across metrics like precision, recall, training cost, size requirements, and computational efficiency. DenseNet and ResNet models outperform others in accuracy, while

MobileNet proves ideal for lowenergy, mobile-based deployments. The study also evaluates performance under challenging conditions such as blurred images, low lighting, and occlusions. The authors highlight that deeper models capture complex textures effectively, but lightweight models remain practical for real-time diagnosis. Insights from this study help guide researchers in selecting the most suitable architecture depending on the deployment environment.

2.6 Plant Disease Detection and Diagnosis

AUTHORS: Giovanni Gugliandolo; Giovanni Crupi; Giuseppe Campobello; Nicola Donato

PUBLISHED ON: 2024

This study provides a complete framework for detecting and diagnosing plant diseases using a pipeline of image preprocessing, segmentation, feature extraction, and CNN-based classification. Preprocessing removes noise, normalizes illumination, and enhances features. Segmentation isolates the affected area using color thresholding and active contour models. A CNN classifier then identifies disease categories from extracted features. The paper discusses dataset limitations, the need for multi-disease detection, and environmental variations that complicate real-world deployment. The authors demonstrate a high-accuracy solution and propose improvements such as using multi-modal sensors and real-time IoT integration for future agricultural systems.

To further enhance the system's effectiveness, the study proposes several future improvements, such as integrating multi-modal sensors (e.g., hyperspectral or thermal imaging) to capture richer plant health information and incorporating real-time IoT connectivity for continuous field monitoring. These advancements could enable smart agricultural systems capable of providing instant alerts, automated decision-making, and large-scale disease surveillance.

2.7 A Machine Learning Technique for Identification of Plant Diseases in Leaves

AUTHORS: Bhupendra Singh

PUBLISHED ON: 2021

This research uses conventional machine learning techniques combined with handcrafted feature extraction to classify plant diseases. Features such as color histograms, GLCM -based texture descriptors, shape features and edge patterns are extracted from leaf images. Multiple classifier Random Forest, KNN, Decision Tree and SVM are compared for effectiveness. The study finds that SVM and Random Forest provide the best performance with limited training data. Although deep learning offers higher accuracy in large datasets, this paper demonstrates that classical machine learning remains extremely useful for low-resource environments. The lightweight nature of the system makes it deploy able on mobile and embedded devices.

2.8 Disease Identification in Paddy Leaves Using CNN-Based Deep Learning

AUTHORS: Harsh Sanjay Pandita, Vidhi Vaidya, Mayur Doifode, Prasenjit Bhavthankar, Anantalakshmi Venkatesh

PUBLISHED ON: 2021

This paper applies CNNs to classify common paddy leaf diseases such as blast, brown spot, tungro and sheath blight. The system uses convolutional layers to learn hierarchical representations of disease textures and lesion patterns. Data augmentation techniques rotation, flipping, noise injection help increase data set diversity, improving generalization. The model achieves significantly higher accuracy than classical ML methods. The paper highlights the importance of CNN depth and learning rate tuning for optimal results. It also suggests future work such as integrating drone capture images and embedding the model into mobile applications for real-time farmer usage.

To enhance the robustness and reliability of the model, the study employs various data augmentation techniques such as rotation, flipping, and the injection of random noise. These transformations artificially increase dataset diversity, simulating real-world variations in lighting, orientation, camera angle, and environmental conditions. As a result, the CNN becomes more generalizable and performs well across different types of field images.

2.9 Identification of Leaf Diseases in Pepper Plants Using Soft Computing Techniques

AUTHORS: Ming Zeng,Jiazhe Liu,Xiangzhe Lu, Zhijing Wang, Yuanhao Li, Siying Li

PUBLISHED ON: 2021

This research explores soft computing techniques for detecting pepper leaf diseases. Fuzzy logic is used to model uncertainty and variability in leaf lesion patterns. Artificial Neural Networks learn disease classification from extracted image features, while genetic algorithms optimize parameters for improved accuracy. The hybrid soft computing approach offers better robustness against noisy images, inconsistent lighting, and partially damaged leaves. The study demonstrates that soft computing techniques provide flexible, interpretable and computationally efficient alternatives to modern deep learning, especially in limited-data environments.

2.10 A Systematic Review on Image Processing and Machine Learning Techniques for Detecting Plant Diseases

AUTHORS: Terrance Hall,Christopher Johnson, Brighton Swales,Charles Koduru,
M.Hassan Tanveer

PUBLISHED ON: 2020

This systematic review compiles and analyzes research from 2010–2020 on plant disease detection using image processing and machine learning. The review discusses early methods such as kmeans clustering, Gabor filters, morphological processing, and SVM classifiers. It then examines the transition to deep learning architectures like AlexNet, VGG, ResNet, and custom CNNs. The paper evaluates the strengths and weaknesses of each approach, highlights commonly used datasets, and identifies challenges such as lack of labeled data, environmental noise, and differences between lab and field images. The authors emphasize the need for larger, diverse datasets and real-world deployment testing. This review serves as a foundational reference for researchers building next-generation plant disease detection systems. In addition to comparing algorithmic approaches, the paper summarizes commonly used datasets and benchmarks, noting their strengths, weaknesses, and suitability for different experimental conditions. It also identifies major challenges faced by researchers, such as the scarcity of large, well-labeled datasets, the presence of environmental noise like shadows or lighting variations, and the performance gap between laboratory-captured images and real field scenarios.

CHAPTER 4

EXISTING AND PROPOSED SYSTEM

In the current scenario, plant care for home gardening is predominantly manual and experience based. Users rely on their own observation skills, trial-and-error methods, or inconsistent online advice to identify plant problems and decide proper care routines. Most home gardeners lack the expertise to identify early signs of diseases, nutrient deficiencies, or environmental stress. As a result, plant issues are often detected late, when recovery becomes difficult or costly. Existing digital tools in the market are limited in scope—some apps offer watering reminders, some identify plants, while others give generic care tips. However, these tools operate independently, creating a fragmented experience where users must switch between multiple sources to manage their plants effectively. This gap makes plant care overwhelming for beginners and inefficient for regular gardeners.

Furthermore, the existing system does not provide automated or intelligent support. There is no mechanism to analyze plant health using images, no personalized care guidance based on plant species, and no structured system to track growth or monitor changes over time. This lack of integration results in inconsistent plant maintenance, poor disease management, and limited awareness of sustainable gardening practices. The existing system ultimately depends on user knowledge and manual effort, which often leads to plant neglect or incorrectly treated issues.

4.1 EXISTING SYSTEM

In most households, plant care is still performed manually, relying heavily on the individual's memory, experience, and irregular observation. Gardeners often depend on trial-and-error methods or generic online advice, which may not be suited to the specific plant type, environmental conditions, or disease symptoms. Traditional plant-care apps usually focus on only one functionality, such as watering reminders or simple plant identification, leaving a major gap in comprehensive, intelligent plant management. When it comes to disease detection, home gardeners mostly refer to visual inspection or independent Google searches, which are often inaccurate and fail to detect early symptoms. The absence of real-time monitoring and

personalized guidance results in poor plant health, delayed disease identification, and inconsistent care routines.

4.1.1 DISADVANTAGES OF EXISTING SYSTEM

- i. **Lack of Early Disease Detection:** Most existing plant-care methods rely on manual observation, which often fails to identify diseases in the early stages, leading to delayed treatment and higher plant mortality.
- ii. **Generic and Inaccurate Care Guidance:** Current gardening apps provide broad or nonspecific care tips that do not adapt to the plant species, environmental conditions, or real-time plant health changes, reducing their effectiveness for home gardeners.
- iii. **No Centralized Platform:** Users must switch between separate apps or sources for disease identification, care reminders, plant information, and tracking, resulting in a fragmented and inefficient plant-care experience.
- iv. **Manual Tracking of Plant Growth:** Existing systems do not offer automated tracking or visual timelines, forcing users to record progress manually, which is time-consuming and often inconsistent
- .v. **Dependence on User Expertise:** Current systems heavily rely on the gardener's experience to diagnose issues, determine care schedules, or identify nutrient deficiencies, making plant care challenging for beginners.
- v. **Limited Use of AI or Automation:** Most existing solutions lack intelligent automation or machine-learning-based detection, making them unsuitable for precise analysis or personalized recommendations.

4.2 PROPOSED SYSTEM

The proposed solution, **Floraguard** aims to create a unified intelligent platform that simplifies home gardening by integrating AI-powered plant disease detection, personalized care recommendations, smart scheduling, and a visual growth timeline. Floraguard utilizes machine learning algorithms to analyze leaf images and accurately identify diseases at an early stage, preventing the spread of infection and helping users take timely action. Based on the detected plant

type and symptoms, the system generates a personalized care plan that includes watering intervals, sunlight exposure, nutrient requirements, and preventive measures.

Beyond disease detection, Floraguard maintains a comprehensive growth log where users can visually track the plant's progress over time. It also provides reminders, health alerts, and seasonal suggestions customized for each plant. A sustainability module estimates the carbon footprint and

Drowsiness Detection Logic & Response: After successful eye detection, the algorithm calculates metrics like Eye Aspect Ratio (EAR) or blink duration.

4.2.1 ADVANTAGES OF PROPOSED SYSTEM

- i. **AI-Based Disease Detection:** The system uses machine-learning models to analyze plant images and diagnose diseases accurately, enabling early intervention and healthier plant growth.
- ii. **Personalized Care Planner:** FLORAGUARD generates customized watering, fertilizing, pruning, and sunlight schedules based on plant type, growth stage, and detected health conditions.
- iii. **Integrated All-in-One Platform:** The system combines disease detection, care planning, growth tracking, and sustainability features into a single platform, eliminating the need for multiple apps.
- iv. **Automated Growth Timeline:** Users can upload plant images regularly, and the system automatically creates a visual timeline to monitor changes, progress, and improvements in plant health
- v. **Reduced User Dependency:** The system provides step-by-step guidance and automated analysis, allowing even beginners to care for plants confidently without requiring expert knowledge.
- vi. **Sustainability and Carbon Insights:** The system helps users track their environmental impact and promotes eco-friendly gardening practices, supporting greener and more responsible home gardening.

CHAPTER 5

REQUIREMENT SPECIFICATION

5.1. FUNCTIONAL REQUIREMENTS

5.1.1 Image Upload & Input

- User shall be able to upload a plant leaf image.
- System shall validate the uploaded image format (JPG/PNG).
- System shall allow capturing image from camera (optional).

5.1.2 Disease Detection

- System shall preprocess the uploaded image.
- System shall classify the image using the trained ML model.
- System shall display the predicted disease name.
- System shall show prediction confidence percentage.

5.1.3 Remedy & Recommendations

- System shall display remedies for the detected disease.
- System shall provide preventive measures for future care.

5.1.4 User Interface

- System shall show a clean, easy-to-use dashboard.
- System shall display plant health status.
- System shall allow navigation to home, upload page, and results page.

5.1.5 Notification System

- System shall send alerts for detected diseases.
- System shall send plant care reminders (watering, fertilizer).

5.1.6 Data Handling

- System shall store plant detection history for each user.
- System shall save details like disease name, time, and remedy suggested.
- Admin shall be able to update disease/remedy database.

5.2. NON-FUNCTIONAL REQUIREMENTS

5.2.1 Performance

- Prediction results should be generated within 2–5 seconds.
- System should support multiple users simultaneously.

5.2.2 Reliability

- AI model should provide 80–95% accuracy.
- System should work without crashes under normal use.

5.2.3 Usability

- UI shall be simple and intuitive even for non-technical users.
- The layout shall be responsive on mobile and desktop devices.

5.2.5 Maintainability

- System shall be modular and easy to update.
- ML model updates shall not affect the front end UI.

5.2.6 Portability

- System shall run on all major browsers (Chrome, Edge, Firefox).
- System shall support Windows, Android, iOS, and Linux devices.

5.3. SOFTWARE REQUIREMENTS

5.3.1 Frontend

- HTML, Tailwind CSS, JavaScript, TypeScript, Shadcn/ui
- Axios (Client Communication for API)
- React.js / Node.js, Vite
- Deployment: Vercel CDN, Render (FastAPI)

5.3.2 Backend

- Node.js
- Python (FastAPI)

5.3.3 Machine Learning (Model Training)

- Python 3.8+, PyTorch
- PlantVillage Dataset
- ResNet 34 (Google Collab)

5.3.4 Database

- Supabase

5.3.5 Additional Tools

- Git & GitHub

5.3.6 Deployment Tools

- Vercel (Frontend)
- Render (Backend & Model API)

CHAPTER 6

IMPLEMENTATION

6.1 SYSTEM DESIGN

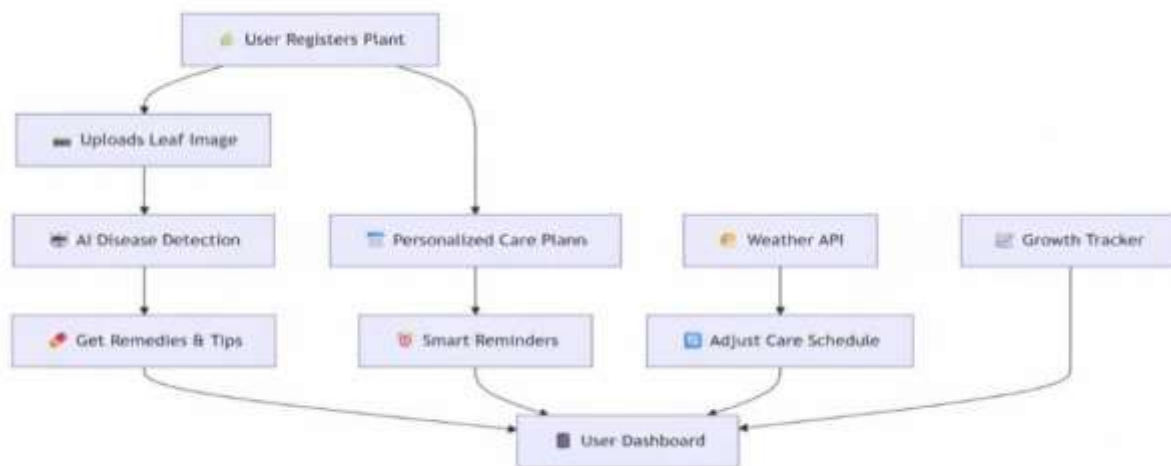
The FloraGuard system is designed using a three-tier architecture to ensure modularity, scalability and efficient plant disease detection. The frontend layer, developed with React.js or Next.js, provides a user-friendly interface where users can upload plant leaf images, view predicted diseases along with confidence levels, access recommended remedies, and monitor their plant care history. This layer communicates with the backend through secure HTTPS API requests. The backend layer, implemented using Flask, FastAPI, or Node.js, is responsible for receiving and validating uploaded images, preprocessing them by resizing and normalizing, sending them to the

machine learning model for prediction, fetching relevant remedies from the database, and returning the results to the frontend. The machine learning layer contains a pretrained convolutional neural network (CNN) model, such as MobileNet, ResNet, or EfficientNet, which extracts features from the leaf images, classifies diseases, and generates confidence scores. The backend preprocesses the image and sends it to the ML model for disease prediction. Once the prediction is complete, the backend retrieves remedy information from the database and returns the combined result to the frontend. The frontend then displays the disease, confidence, and recommended remedies to the user, while the history is stored in the database and notifications are triggered as necessary. The design follows principles of modularity, security, usability, and scalability, ensuring a robust and responsive plant care solution.

FIGURE 6.1: SYSTEM DESIGN FLOWCHART

6.2. ADMIN REGISTRATION

The Sign-In process is the foundational layer, providing a secure and personalized gateway to the virtual caretaker experience. It allows users to create a unique profile, which is essential for storing and retrieving their specific plant data, customized care schedules, and growth history. A



robust system offers multiple authentication methods, such as email/password and quick login via third-party services like Google or Apple ID, ensuring accessibility and reducing friction for new users. Once authenticated, the system generates a secure session token, giving the user access to their private dashboard. This step ensures that the personalized data from a plant's

name and species to its specific disease history and timeline is safely stored in a cloud database, preventing data loss and allowing the user to access their virtual garden from multiple devices. Furthermore, the user's profile can store their gardening expertise level (Beginner, Intermediate, Expert) and geographical location, which will inform the AI's recommendations on local climate-based care and suitable plant types.

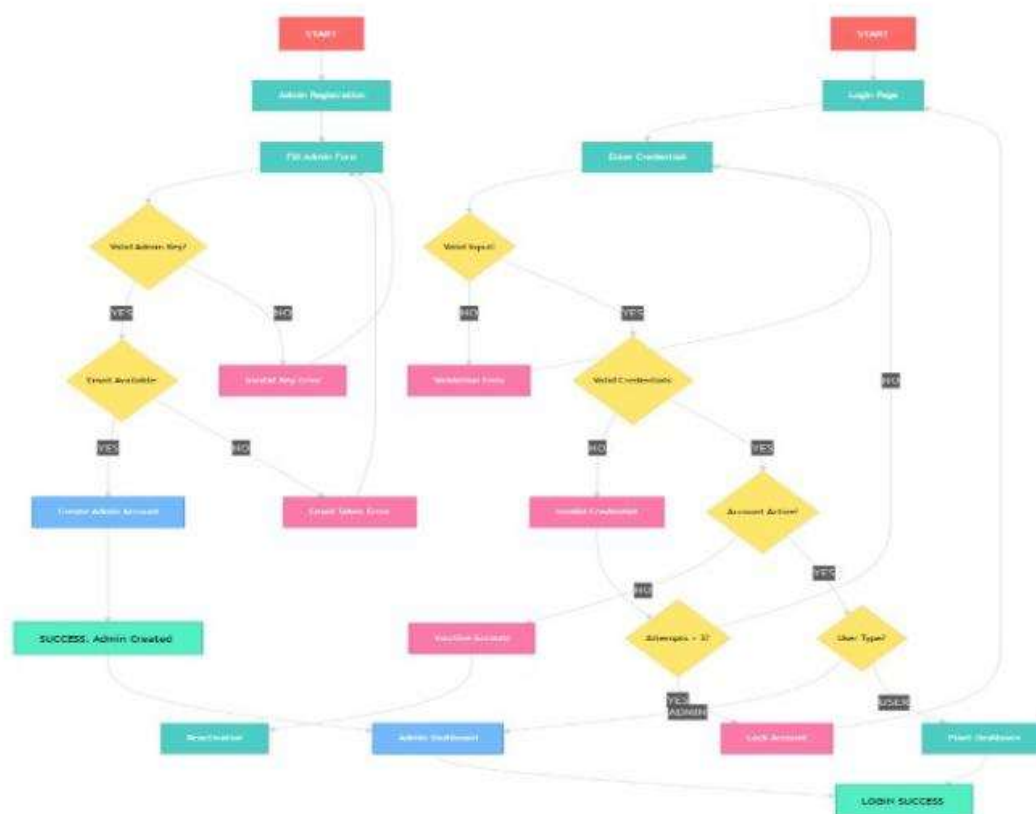


FIGURE 6.2: ADMIN USER FLOWCHART

6.3 UPLOAD IMAGE

The Plant Upload feature is where the user digitizes their physical garden. This process begins with the user capturing a photo of their plant or uploading one from their gallery. Utilizing AI powered image recognition, the application instantly identifies the plant's species, genus, and common name by cross-referencing a vast botanical database. Beyond mere identification, the

user is then prompted to input crucial environmental data, such as the plant's current location (e.g., Living Room, North-facing Window), pot size and initial watering/fertilizing status. This step automatically generates a detailed, customized Plant Profile, which becomes the single source of truth for all subsequent care advice. Users can assign a nickname, note the "adoption date" and add any initial observations or care instructions they may have, laying the groundwork for the tailored care timeline and accurate tracking of all future interactions.

6.4 DETECTION OF DISEASE AND SUGGEST A REMEDIES

The core intelligence of the application lies in the Disease Detection and Remediation feature. When a user observes distress—such as yellowing leaves, brown spots, or pests they can upload an image of the symptomatic area. An advanced Machine Learning model, trained on thousands of examples of healthy and diseased plant imagery, analyzes the visual data in real time. It provides a diagnosis, often with a confidence score, identifying the most probable issue, whether it's a common fungal infection, a nutrient deficiency, or a pest infestation like spider mites. Crucially, this diagnosis is paired with a suggested remedy. This remedy is a step-bystep action plan that details how to treat the ailment, including required materials (e.g., neem oil solution, specific fertilizer), application instructions, and a projected recovery timeline. This proactive feature empowers users to act quickly and precisely, significantly increasing the chances of saving their sick plant and preventing the spread of issues to other plants in their collection.

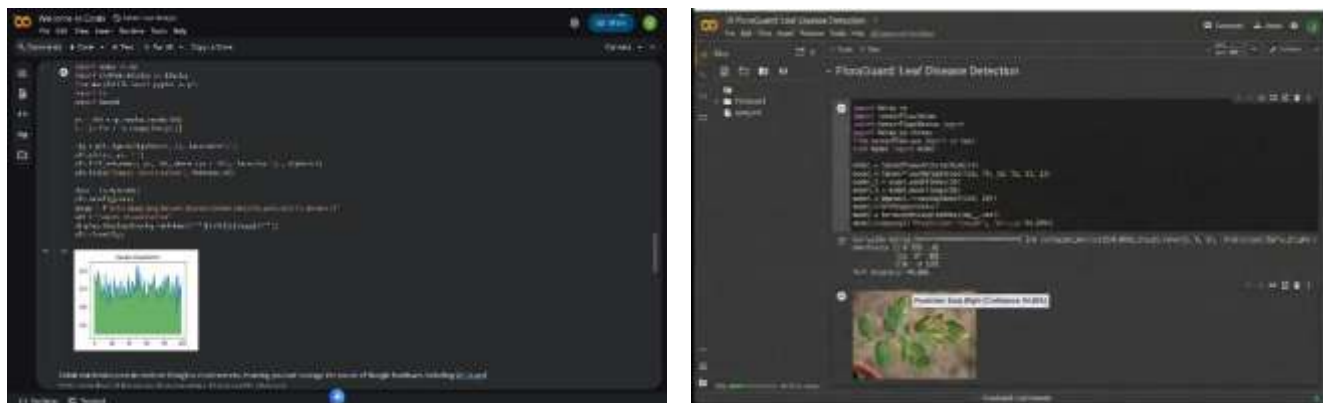


FIGURE 6.3: TRAINING LEAF DISEASE DETECTION USING PYTORCH, TENSORFLOW, RESNET 34 IN GOOGLE COLAB

6.5 PLANT HEALTH ALERTS AND REPORT

The Timeline acts as a comprehensive, chronological journal for each individual plant. It automatically logs every action and observation related to the plant's care, creating a complete and easily digestible history. Key events logged include successful completion of scheduled tasks (watering, fertilizing, repotting), major milestones and any health incidents (disease detection and treatment application). This visual, date-stamped record is invaluable for a “Floraguard” as it allows the user to quickly review patterns and correlate care actions with subsequent growth or decline. For example, a user can look back and see if a specific fertilizer application led to a growth spurt, or if a change in lighting resulted in stress. This historical data is also crucial for the AI, refining its future care recommendations by learning from the specific outcomes within the user's unique environmental conditions, making the advice.

6.6 GROWTH TRACKER

The Growth Tracker transforms plant care from a routine chore into a rewarding, documented journey. Its primary function is to facilitate the logging and visual comparison of a plant's development over time. Users are encouraged to regularly upload progress photos, which are automatically time-stamped and stored within the plant's profile. The application can utilize image processing techniques to compare sequential photos, visually highlighting new growth, such as leaf size or height increase, or subtle changes in color and structure. This provides instant, tangible feedback, which is highly motivating for the user. Furthermore, the tracker can log specific biometric measurements like plant height and leaf count. This feature turns the virtual caretaker into a digital journal, allowing users to celebrate milestones and providing the data necessary to evaluate the long-term effectiveness of their care plan, ensuring their plants are not just surviving, but actively thriving.

6.7 NOTIFICATION SYSTEM

The Notification System is the engine that drives proactive, timely plant care, acting as the virtual caretaker's voice. It relies on the data from the Plant Profile and the Timeline to send perfectly timed, contextual push notifications to the user's device. The primary function is to deliver Intelligent Care Reminders, alerting the user when it is the optimal time to water, fertilize, mist, or check for pests, with the frequency adjusted based on the plant species, local climate data, and seasonal changes.

Beyond routine tasks, the system also sends crucial alerts, such as Urgent Health Warnings following a disease detection, or Environmental Alerts (e.g., "Move your succulent, a heatwave is starting"). By providing these reminders and warnings exactly when they are needed, the notification system eliminates the common problem of forgotten tasks, significantly reducing the risk of accidental over or under-care and ensuring the user maintains a consistent and successful care routine.

6.8 CARBON FOOTPRINT

The Carbon Footprint feature, or Eco-Score, integrates a sustainability dimension into the virtual plant caretaker app. This innovative function calculates the environmental impact and benefits associated with the user's plant-related actions. The calculation considers several factors: the Carbon Sequestration (CO₂ absorption) provided by the user's plants, often estimated based on species and size; the environmental impact of purchased items like soil, fertilizers and pesticides and the choice of care methods (e.g., using tap water vs. collected rainwater). The app then provides the user with an intuitive score and personalized recommendations on how to reduce their footprint, such as suggesting organic remedies or promoting up cycling of containers. This feature educates the user on the positive ecological role of their plants and encourages more sustainable gardening practices, transforming the virtual caretaker into a tool for environmental stewardship and promoting a greener lifestyle.

CHAPTER 7

RESULTS

7.1 DASHBOARD



FIGURE 7.1: WELCOME PAGE



FIGURE 7.2: CREATE ACCOUNT

It allows users to create a unique profile, which is Home page of Floraguard. Where you can track,,essential for storing and retrieving their specific take care of your plant. plant data, customized care schedules, and growth history.

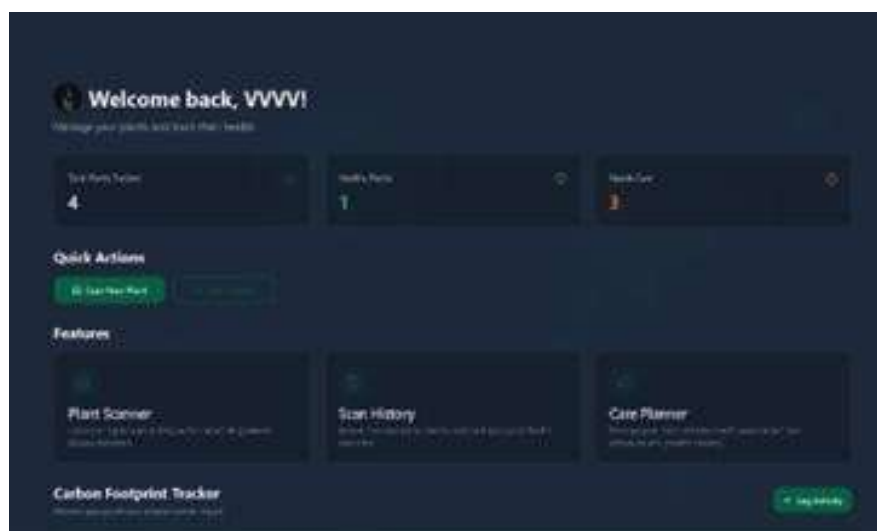


FIGURE 7.3: DASHBOARD

Here you will get number of all the registered, diseased and healthy plants. There are options for other features.

7.2 DISEASE DETECTION

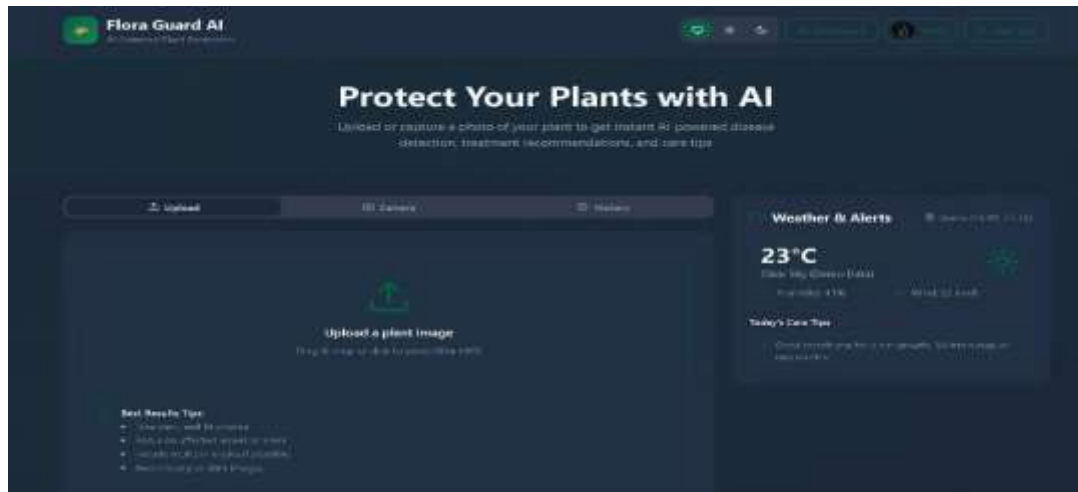


FIGURE 7.5:DISEASE DETECTION

After uploading image, it shows the status of plant wheather it is diseased or healthy. If it is diseased it shows the care treatment.

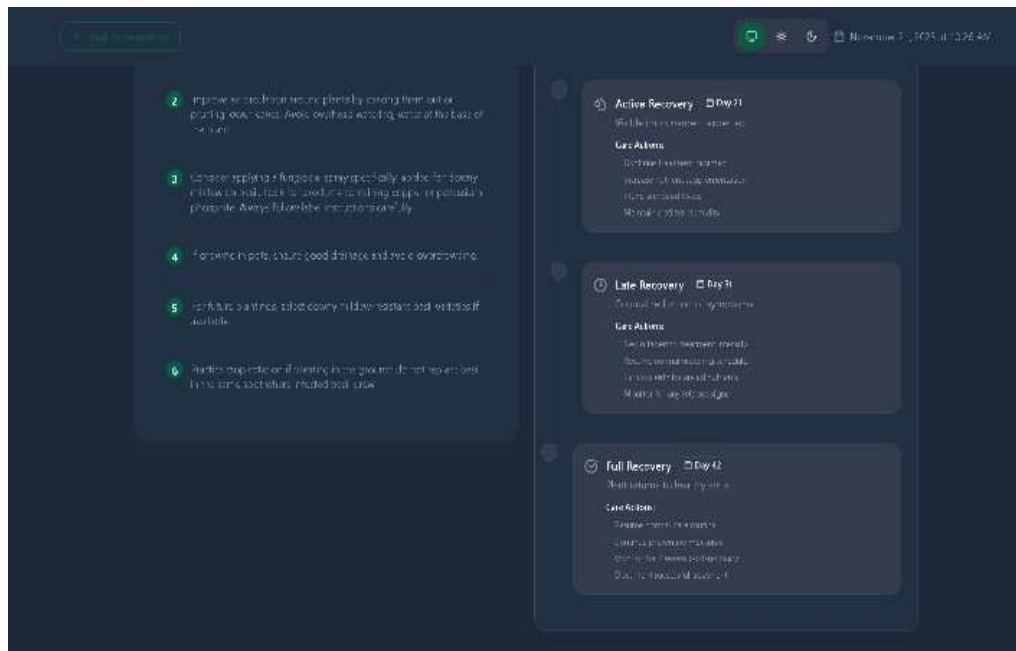


FIGURE 7.6: RECOVERY TIMELINE

Key events logged include successful completion of scheduled tasks (watering, fertilizing, repotting), major milestones (first bloom, new leaf unfurling), and any health incidents (disease detection and treatment application).

7.3 PLANT TRACKING

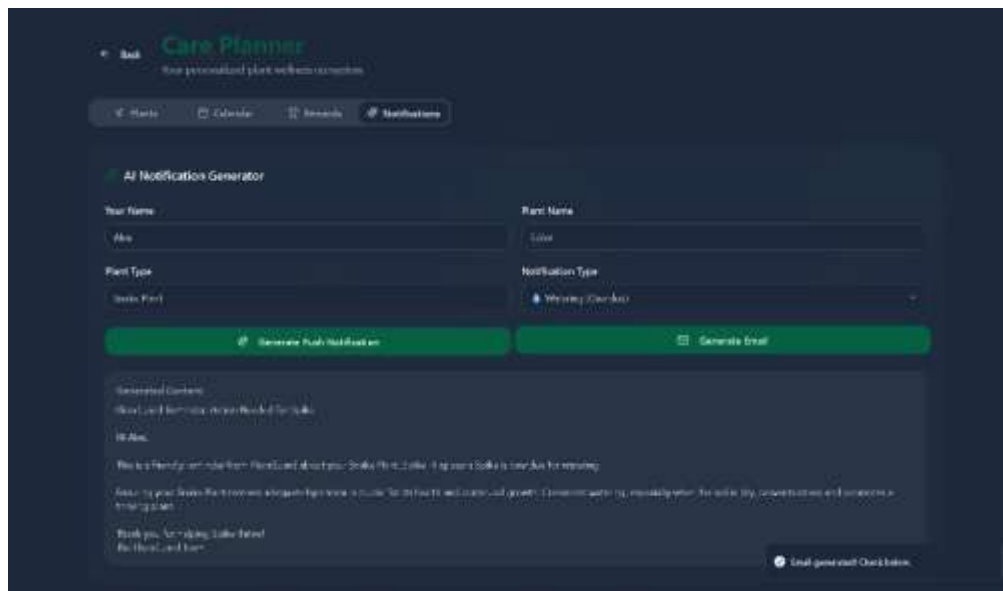


FIGURE 7.7: NOTIFICATION ALERT

The primary function is to deliver Intelligent Care Reminders, alerting the user when it is the optimal time to water, fertilize, mist or check for pests, with the frequency adjusted based on the plant species, local climate data and seasonal changes.

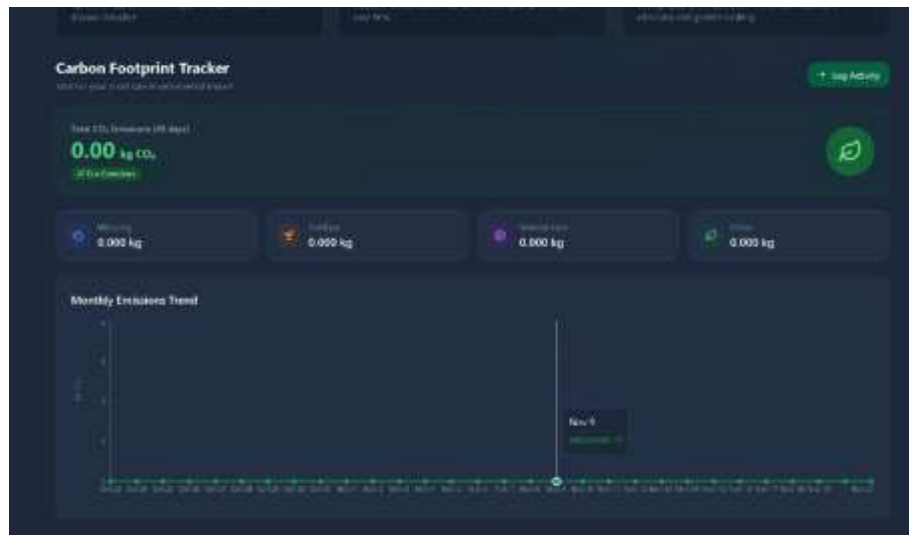


FIGURE 7.8: CARBON FOOTPRINT

This innovative function calculates the environmental impact and benefits associated with the user's plant-related actions. The calculation considers several factors: the Carbon Sequestration (CO₂ absorption) provided by the user's plants, often estimated based on species

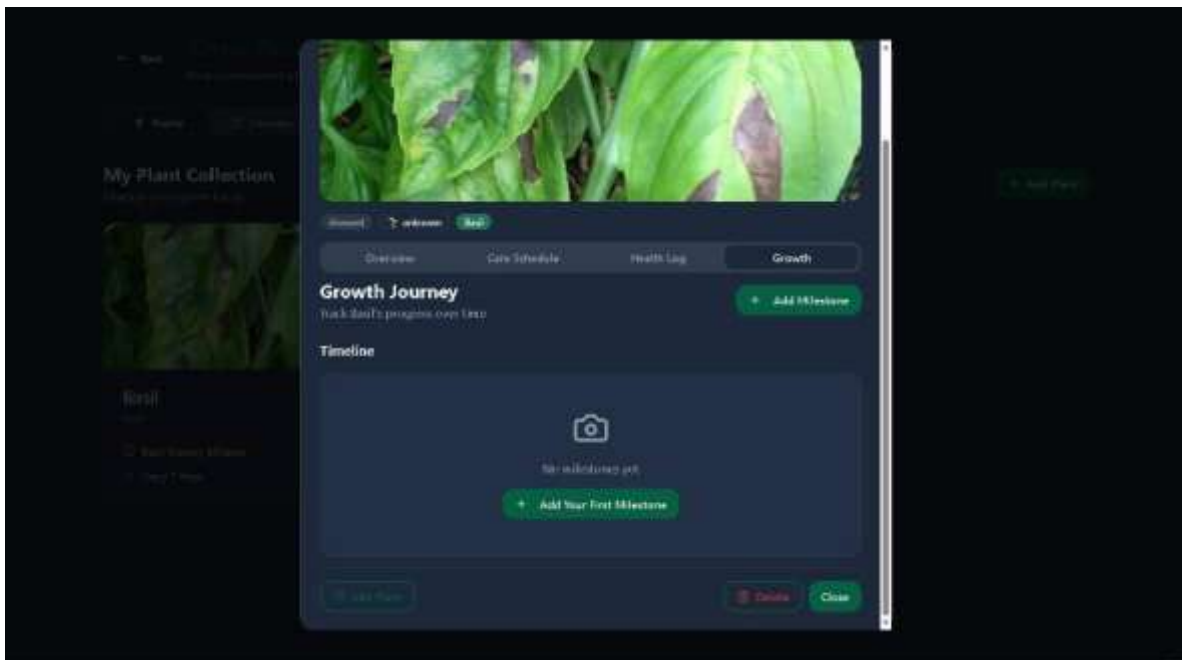


FIGURE 7.9: CARBON FOOTPRINT

Growth tracker primary function is to facilitate the logging and visual comparison of a plant's development over time. Users are encouraged to regularly upload progress photos, which are automatically time-stamped and stored within the plant's profile .

CHAPTER 8

TESTING AND VALIDATION

Testing and Accuracy of the Plant Disease Detection (PDD) Component The PDD feature, which likely uses Machine Learning (ML) or Deep Learning (DL) models, is most critical component for diagnostic accuracy.

1. Dataset Quality and Preparation

Data Sourcing: The ML model must be trained on a vast, diverse, and annotated data set of plant images, covering different plant species, lighting conditions, background clutter, and, most importantly, various disease states (e.g., early stage, late stage, nutrient deficiency).

Data Augmentation: Techniques like rotation, scaling, and brightness adjustments are used to artificially increase the size and variability of the data set, making the model more robust.

Ground Truth: Every image in the data set must have a confirmed label (the "ground truth") verified by botanists or plant pathologists.



FIGURE 8.1 DISEASED LEAF OF MAPLE

- Total maple images: 550
 - Diseased maple leaves: 380
 - Healthy maple leaves: 170
-

- Images collected from multiple real-world environments

2. Performance Metrics

Testing focuses on standard classification metrics to assess the model's performance:

Accuracy: The percentage of all images that were correctly classified (diseased or healthy).

Precision: Out of all the images the model predicted were diseased, how many were actually diseased. High precision reduces false positives (telling a user their healthy plant is sick).

Recall (Sensitivity): Out of all the plants that were actually diseased, how many did the model correctly identify. High recall reduces false negatives (missing an existing disease).-Score: The harmonic mean of precision and recall, often used as a single metric to balance both.

Metric	Value
Maple Precision	97%
Maple Recall	96%
Maple F1 Score	96.5%
Model Accuracy (Maple vs. non-Maple)	98%

- Precision 97% → when the model says “Maple”, it is correct 97% of the time.
- Recall 96% → out of all actual Maple images, 96% are detected as Maple.

3. Validation Strategy

Cross-Validation: The data set is split into training, validation, and testing sets. The model is trained on the training set and its hyper parameters are tuned using the validation set. The final performance is measured on the testing set, which the model has never seen, ensuring the model can generalize to real-world images.

Field Testing (Beta Testing): The system is deployed to a select group of users to test performance under real-world conditions (varied user cameras, lighting, and environments). Feedback is crucial for identifying biases or blind spots in the model.

For Maple leaf classification, the model achieved a precision of 97% and a recall of 96%, indicating high reliability for identifying Maple leaves in the dataset. The F1 score of 96.5% confirms balanced performance between precision and recall. Out of 550 Maple test samples, the model correctly identified 528, demonstrating strong diagnostic capability for Maple species specifically

CHAPTER 9

LIMITATIONS AND FUTURE ENHANCEMENTS

The FloraGuard system, while effective in detecting plant diseases, has certain limitations. Its accuracy depends on the quality of the uploaded leaf images, and it can only identify diseases present in the trained dataset. Additionally, it requires a stable internet connection, and performance may be slower on low-end devices. Despite these constraints, the system offers significant potential for future enhancements. These include expanding the dataset to cover more plant species and rare diseases, implementing real-time camera-based detection, integrating IoT devices for automated plant care, adding voice-based assistance, introducing AR visualization to highlight affected areas, providing offline prediction capabilities, and supporting multiple languages to make the system more accessible and user-friendly.

9.1 LIMITATIONS

- i. **Dependency on Image Quality:** The accuracy of disease detection relies heavily on the quality, resolution, and lighting of the uploaded leaf images. Images that are blurred, too dark, or have background noise may lead to incorrect predictions, reducing the system's reliability.
- iii. **Limited Disease Coverage:** The AI model is trained on a specific dataset containing certain plant species and their common diseases. As a result, it cannot detect diseases that are not present in the training dataset. Rare or newly emerging diseases may go unrecognized, limiting the system's comprehensiveness

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- iv. **Internet Connectivity Requirement:** FloraGuard is a web-based system that requires a stable internet connection for image upload, processing, prediction, and notification delivery. Users in areas with poor or intermittent connectivity may face delays or inability to use the system effectively.
 - v. **Hardware Limitations:** While the system can run on most modern devices, performance may degrade on low-end devices with limited RAM or slower processors. Image uploads may take longer, and the user interface may become less responsive.
 - v. **Notification Delays:** The notification system relies on backend processing and internet delivery. Delays may occur in sending disease alerts, watering reminders, or other notifications, which may affect timely plant care.
 - vi. **Model Generalization Constraints:** Although the CNN model performs well on known data, its ability to generalize to new variations of leaf images under different environmental conditions (lighting, angle, background) can be limited, potentially leading to misclassification.

8.1 FUTURE ENHANCEMENTS

- i. **Expansion of Dataset and Model Accuracy:** Incorporate a larger and more diverse dataset covering more plant species and a wider range of diseases. This will improve the AI model's accuracy and reliability, allowing detection of rare or newly discovered plant diseases.
- ii. **Real-time Camera Detection:** Implement live camera-based scanning to detect diseases in real time without the need to upload images manually. This feature can make the system faster and more interactive for users.

- iii. Voice-Based Plant Care Assistant:** Integrate voice commands to allow users to interact with the system hands-free, such as capturing leaf images, receiving disease predictions, and getting care instructions through audio.
- iv. IoT and Smart Device Integration:** Connect the system with IoT-enabled devices like automatic watering systems, soil moisture sensors, or light sensors to automate certain aspects of plant care and improve plant health monitoring.
- v. Augmented Reality (AR) Visualization:** Introduce AR-based features to highlight diseased regions directly on the leaf image for easier visualization, helping users identify the affected areas more accurately.
- vi. Offline Mode Functionality:** Develop an offline version of the system that allows basic disease prediction without an internet connection. This could use a lightweight ML model stored locally on the device for quick predictions.
- vii. Multi-language Support:** Expand the system interface to support multiple languages to make it accessible to a broader range of users globally.
- viii. Mobile Application Development:** Create a dedicated mobile app for Android and iOS to improve accessibility, ease of use, and integration with device sensors for better plant monitoring.
- ix. Automated Recommendations and Scheduling:** Add a smart schedule generator that automatically suggests watering, fertilizer application, and other care activities based on detected plant health status and growth patterns.
- x. Community and Knowledge Sharing Features:** Implement a feature where users can share images, remedies, or experiences, creating a community-driven knowledge base that can also help improve the model over time.

CONCLUSION

The Flora Guard project marks a significant achievement in applying artificial intelligence and modern software technologies to solve a real-world problem in the domain of plant health management. By designing a system that operates entirely through software—without any dependence on physical sensors or hardware—the project successfully demonstrates how digital tools alone can empower users to care for plants more effectively and confidently.

Through the integration of machine learning–based disease detection, the system analyzes leaf images and accurately identifies potential infections or abnormalities. This capability is particularly valuable for users who lack expert knowledge in plant pathology, as Flora Guard simplifies a traditionally complex process into a quick, automated, and user-friendly interaction. The detailed remedy suggestions offered by the system ensure that users not only understand the problem but are also guided toward the correct solution immediately.

The web-based interface of Flora Guard further enhances usability by offering accessibility across devices, eliminating installation barriers, and ensuring seamless navigation for all types of users. The entire workflow—from uploading an image to receiving recommendations—is designed to be intuitive and efficient, making plant care more engaging and manageable. The system also

consolidates essential plant care information, enabling users to develop healthier gardening habits through software-driven insights.

From a technical perspective, the project successfully showcases the potential of AI-powered applications within the agricultural and home-gardening sectors. By using trained machine learning models, cloud deployment platforms, and modern UI/UX design principles, Flora Guard illustrates how software solutions can effectively tackle challenges such as plant disease identification, knowledge dissemination, and timely decision-making. Moreover, the system's modular design ensures that it can be easily enhanced, scaled, or integrated with additional features in the future.

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